Conceptual Design of Aircraft
Jet engine – Further examples

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1. A by-pass engine is flying at a speed of 100 m/s in standard sea level conditions. Its by-pass ratio is 4 and fuel to air ratio is 0.01. The cold air leaves the engine at a speed of 200 m/s and the hot gas leaves the engine at a speed of 300 m/s.

   a) Calculate the specific net thrust, specific fuel consumption (sfc) and propulsive efficiency.

   b) You are asked to reduce the sfc at these flight conditions by 50%. Assuming that the hot- and cold-stream exit velocities remain unchanged, estimate the required by-pass ratio. Calculate the corresponding specific net thrust and propulsive efficiency.

2. Using the scaling laws for jet engines, estimate the sfc at cruise in Question 1 for a by-pass ratio of 4. Compare the result with Question 1 result and discuss the difference.

3. The by-pass engine in Question 2 was fitted with a re-heat system. Using the scaling laws for jet engines, estimate the sfc at cruise with a re-heat system.

4. Using the scaling laws for jet engines, derive a scaling law for the engine's mean mass density as a function of the ratio between the required thrust and the known thrust.

Standard sea level conditions:

$\rho = 1.225$ kg/m^3, $p = 101.325$ kPa, $T = 288.16$ K, $c = 340.3$ m/s

1. By-pass engine

Data: V_a = 100 m/s, μ = 4, f = 0.01, V_{jh} = 300 m/s, V_jc = 200 m/s.

a) Specific net thrust, sfc and propulsive efficiency

The net thrust is:

$$F_n = \dot{m}_{jh} V_{jh} + \dot{m}_{jc} V_{jc} - \dot{m}_a V_a$$

$$\frac{F_n}{\dot{m}_a} = \frac{\dot{m}_{jh}}{\dot{m}_a} V_{jh} + \frac{\dot{m}_{jc}}{\dot{m}a} V_{jc} - V_a$$

$$\frac{F_n}{\dot{m}_a} = \frac{\dot{m}_{jh}}{\dot{m}_{jh} + \dot{m}_{jc}} V_{jh} + \frac{\dot{m}_{jc}}{\dot{m}_{jh} + \dot{m}_{jc}} V_{jc} - V_a$$

$$SNT = \frac{V_{jh}}{1+\mu} + \frac{\mu V_{jc}}{1+\mu} - V_a$$

since $\mu = \dot{m}_{jc} / \dot{m}_{jh}$

$$SNT = \frac{300}{1+4} + \frac{4 \times 200}{1+4} - 100$$

$$SNT = 120 \quad \frac{N}{kg/s}$$

Specific fuel consumption:

$$sfc = \frac{\dot{m}_f}{F_n} = \frac{f \dot{m}_{jh}}{F_n} = \frac{f \dot{m}_a}{F_n (1+\mu)}$$

$$sfc = \frac{f}{(1+\mu) SNT}$$

$$sfc = \frac{0.01}{(1+4) \times 120} \times 3600 = 0.0600 \quad \frac{kg}{N \cdot hr}$$

Propulsive efficiency:

$$eta_p = \frac{F_n V_a}{F_n V_a + \frac{\dot{m}_{jh} (V_{jh} - V_a)^2}{2} + \frac{\dot{m}_{jc} (V_{jc} - V_a)^2}{2}}$$

$$\eta_p = \frac{\frac{F_n V_a}{\dot{m}_{jh}}}{\frac{F_n V_a}{\dot{m}_{jh}} + \frac{(V_{jh} - V_a)^2}{2} + \frac{\mu (V_{jc} - V_a)^2}{2}}$$

$$\dot{m}_a = \dot{m}_{jh} + \dot{m}_{jc} = \dot{m}_{jh}(1+\mu)$$

$$\dot{m}_{jh} = \frac{\dot{m}_a}{1+\mu}$$

$$SNT = \frac{F_n}{\dot{m}_a} = \frac{F_n}{\dot{m}_{jh}(1+\mu)}$$

$$\eta_p = \frac{SNT(1+\mu)V_a}{SNT(1+\mu)V_a + \frac{(V_{jh}-V_a)^2}{2} + \frac{\mu(V_{jc}-V_a)^2}{2}}$$

$$\eta_p = \frac{120(1+4)\times 100}{120(1+4)\times 100 + \frac{(300-100)^2}{2} + \frac{4(200-100)^2}{2}} = 0.6$$

1(b). Reduce sfc by 50%

The hot- and cold-stream exit velocities remain unchanged, but the specific net thrust changes with by-pass ratio.

$$SNT = \frac{V_{jh} + \mu V_{jc}}{1+\mu} - V_a$$

$$sfc = \frac{f}{(1+\mu)SNT} \times 3600$$

Combining these equations gives:

$$(1+\mu)SNT = V_{jh} + \mu V_{jc} - (1+\mu)V_a$$

$$(1+\mu)SNT = 300 + 200\mu - 100(1+\mu) = 200 + 100\mu$$

$$sfc = \frac{0.01\times 3600}{200+100\mu} = \frac{0.36}{\mu+2}$$

From part (a):

$$sfc_{old} = 0.0600 \ \ \frac{kg}{N\cdot hr}$$

$$sfc_{new} = 0.5\times 0.0600 = 0.0300 \ \ \frac{kg}{N\cdot hr}$$

$$0.0300 = \frac{0.36}{\mu_{new}+2}$$

$$\mu_{new} = 10$$

The corresponding specific net thrust is:

$$SNT_{new} = \frac{300 + 10 \times 200}{1 + 10} - 100$$

$$SNT_{new} = 109.1 \ \frac{N}{kg/s}$$

The corresponding propulsive efficiency is:

$$\eta_p = \frac{SNT(1+\mu)V_a}{SNT(1+\mu)V_a + \frac{(V_{jh} - V_a)^2}{2} + \frac{\mu(V_{jc} - V_a)^2}{2}}$$

$$\eta_p = \frac{109.1(1+10) \times 100}{109.1(1+10) \times 100 + \frac{(300-100)^2}{2} + \frac{10(200-100)^2}{2}}$$

$$\eta_p = 0.632$$

2. Scaling-law estimate of cruise sfc

Data: μ = 4.

$$sfc_{cruise} = 0.88\, e^{-0.05\mu} = 0.88\, e^{-0.05 \times 4}$$

$$sfc_{cruise} = 0.72048 \ \frac{lb/hr}{lb} = 0.72048 \ \frac{kg/hr}{kg}$$

$$sfc_{cruise} = 0.0735 \ \frac{kg}{N \cdot hr}$$

From Question 1:

$$sfc = 0.0600 \ \frac{kg}{N \cdot hr}$$

The scaling-law estimate is about 22.4% higher than the value from Question 1.

This is caused by the scaling law, which is an average of various engines. Hence the calculation in Question 1 should be seen as more accurate.

3. Scaling-law estimate with re-heat

Data: μ = 4, re-heat system.

$$sfc_{cruise} = 1.04\, e^{-0.188\mu} = 0.4903 \ \frac{lb/hr}{lb}$$

$$sfc_{cruise} = 0.0500 \ \frac{kg}{N \cdot hr}$$

Note that sfc at cruise actually decreased.

According to the scaling law, sfc at cruise decreases due to the re-heat system when:

$$1.04\,e^{-0.188\mu} = 0.88\,e^{-0.05\mu}$$

$$\mu > 1.21$$

This does not happen at take off, where a re-heat system will always increase sfc.

The following explanatory phrase is partly obscured in the handwritten original: it refers to the scaling of sfc at take off and to F_n at cruise.

4. Mean mass-density scaling law

Let SF denote the ratio between the required thrust and the known-data thrust.

$$\rho = \frac{m}{V} \approx \frac{W}{V}$$

The jet-engine volume is modelled as a circular cylinder:

$$V \approx D^2 L$$

$$\frac{\rho_{req}}{\rho_{data}} = \frac{\frac{W_{req}}{W_{data}}}{\left(\frac{D_{req}}{D_{data}}\right)^2 \left(\frac{L_{req}}{L_{data}}\right)}$$

Using the scaling laws in the handwritten solution:

$$\frac{\rho_{req}}{\rho_{data}} = \frac{(SF)^{1.1}}{(SF)^{0.8}(SF)^{0.4}}$$

$$\frac{\rho_{req}}{\rho_{data}} = (SF)^{-0.1}$$

Thus, as the engine size/thrust scaling factor increases, the mean density decreases.